Brain Tumor Segmentation using Attention-Based Multi-Scale U-Net with Residual Learning, FPN, and Hybrid Loss Optimization

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***Abstract*—Segmenting brain tumors using Magnetic Resonance Imaging (MRI) is crucial for computer-aided diagnosis as it aids in treatment planning, surgical guidance, and disease monitoring. Manual annotation by radiologists is accurate mostly but time-consuming. This research introduces an improved deep learning system for brain tumor segmentation using the U-Net architecture. It includes attention gates, residual blocks, a Feature Pyramid Network (FPN), pyramid pooling, and a hybrid loss formulation. The suggested model addresses weaknesses of baseline methods, including missing tiny tumors, inadequate border reconstruction, and false positive predictions in healthy tissue. The LGG MRI Segmentation dataset was used for experiments with PyTorch on Google Colab. The system successfully segmented with high accuracy (0.995), dice score (0.839), sensitivity (0.781), and specificity (0.997). Using stable feature learning, multi-level feature fusion, multi-scale context extraction, and attention-guided decoding leads to improved robustness and segmentation quality in medical imaging applications.**

***Index Terms*—Brain tumor segmentation, MRI, U-Net, Attention U-Net, residual learning, Feature Pyramid Network, pyramid pooling, deep learning, medical image analysis.**

I. Introduction

Brain tumors are abnormal cell growths that can lead to death if not found early. Accurately identifying tumor boundaries is crucial for diagnosis, treatment planning, radiation, and ongoing monitoring. MRI is a popular imaging modality because to its ability to give high-quality soft-tissue contrast without exposing patients to radiation. [1].

Traditionally, radiologists manually review MRI scans and identify worrisome areas. Although effective, this procedure needs knowledge, takes time, and may differ among observers. Automated segmentation systems can save burden, enhance uniformity, and let clinicians make faster decisions.

Segmentation predicts the specific tumor territory pixel by pixel, while classification just determines its presence. This increases clinical utility while also increasing technical demands. This research built and enhanced a deep learning-based segmentation pipeline to improve localization, bounds, and reduce false alarms compared to the standard U-Net.

The rest of the paper is organized as follows: Section II states the problem; Section III reviews related work; Section IV describes the proposed methodology and architecture; Section VI presents experimental results; Section VII provides a comparative evaluation; and Section X concludes with directions for future work.

II. Problem Statement

While convolutional neural networks and traditional U-Net models excel in certain settings, they nevertheless have practical limits in brain tumor segmentation:

- Small tumors are typically overlooked due to restricted receptive fields.
- Tumor boundaries may appear blurry or partial in reconstructed masks.
- Healthy tissue may be mislabeled as a tumor, leading to an increase in false positives.
- Tumors can vary widely in shape, size, and texture amongst persons.
- Medical datasets are significantly smaller than standard computer vision datasets.
- Tumor pixels occupy a limited area, resulting in a large class imbalance.

This study aimed to provide a more effective segmentation framework that enhances detection accuracy, reduces false positives, and preserves precise structural details of tumor boundaries.

III. Literature Review

A. U-Net

U-Net [1] is a popular architecture for segmenting biomedical images. The encoder-decoder design uses skip connections to transport spatial information between encoding and decoding layers, preserving fine-grained detail during mask reconstruction.

B. Attention U-Net

Attention U-Net [2] inserts attention gates into skip paths. These modules enable the network to focus on tumor-relevant regions while suppressing irrelevant background responses, resulting in clearer forecasts and fewer false alarms.

C. Pyramid Pooling

Pyramid Scene Parsing Network (PSPNet) [3] proposes pyramid pooling which captures contextual information across several receptive fields by combining features at different scales. This technique enhances segmentation for both small and big tumor locations.

D. Residual Learning

Residual connections [4] improve gradient flow in deep networks. Instead of learning a whole transformation, each residual block learns a refinement over the identity mapping, resulting in stable optimization and faster convergence.

E. Feature Pyramid Network

FPN [5] combines information from several encoder depths by top-down upsampling and lateral connections. Shallow layers preserve boundaries and geographic detail, whereas deeper layers offer semantic comprehension. Their combination enhances localization performance across different object scales.

F. Multi-Scale Feature Fusion

Recent study indicates that incorporating contextual information with hierarchical features improves segmentation efficiency, particularly in medical imaging where diseases fluctuate in size and appearance.

IV. Proposed Methodology

The entire process of the proposed system is as follows:

$$\text{MRI Image} \rightarrow \text{Preprocessing} \rightarrow \text{Improved Network} \rightarrow \text{Predicted Mask} \rightarrow \text{Evaluation}$$

The proposed framework expands basic U-Net with five major enhancements: (1) residual blocks for stable training, (2) a basic FPN for multi-level feature fusion, (3) attention gates for selective feature routing, (4) pyramid pooling for multi-scale context extraction, and (5) a hybrid loss function for improved overlap and hard-pixel learning. This combination addresses optimization stability, representation quality, scale variation, and false positive suppression into a unified pipeline.

A. Dataset

The project utilizes the **LGG MRI Segmentation Dataset** [6], publicly available on Kaggle. The dataset includes MRI slices and tumor masks for low-grade glioma (LGG) cases. Table I highlights the dataset's characteristics.

TABLE I
LGG MRI Segmentation Dataset Summary

Parameter	Description
Dataset Name	LGG MRI Segmentation
Imaging Modality	MRI
Annotation Type	Pixel-wise Tumor Masks
Tumor Category	Low-Grade Glioma
Source	Kaggle
Learning Type	Supervised Segmentation

B. Preprocessing

Prior to network training, medical pictures must be carefully preprocessed. The following steps were taken for all image-mask pairs:

1) **Grayscale Conversion:** Images were changed to grayscale to save complexity and memory usage.
2) **Resizing:** All images and masks were resized to 256×256 pixels for uniform batch processing.
3) **Normalization:** Pixel intensity values have been scaled to the range. $[0, 1]$.
4) **Empty Slice Removal:** Slices lacking tumor areas were screened to improve training efficiency and eliminate class imbalance.
5) **Data Split:** The dataset was divided into 80% training and 20% validation subsets.

C. System Architecture

The encoder uses a conventional contracting approach that includes convolution and max-pooling, with residual blocks merged at each stage. A basic FPN combines feature maps from many encoder levels via top-down upsampling and lateral connections. The bottleneck uses a pyramid pooling module to capture local and global context simultaneously. Skip connections with attention gates choose tumor-relevant features for decoding. The decoder upsamples fused feature maps to recreate the segmentation mask, culminating in a 1×1 convolution followed by sigmoid activation for binary output. Fig. **??** illustrates the high-level system architecture.

D. Residual Blocks

Each encoder stage includes residual learning:

$$\mathbf{y} = \mathbf{x} + \mathcal{F}(\mathbf{x}) \tag{1}$$

where $\mathbf{x}$ is the block input, $\mathcal{F}(\mathbf{x})$ represents the residual mapping learned by two convolutional layers with batch normalization, and $\mathbf{y}$ is the block output. This formulation enables direct gradient flow and prevents feature deterioration in deeper structures.

E. Feature Pyramid Network

Encoder feature maps at depths $\{C_2, C_3, C_4, C_5\}$ are combined top-down to produce a set of pyramid feature maps $\{P_2, P_3, P_4, P_5\}$. Each level is computed as:

$$P_i = \text{Conv}_{1\times1}(C_i) + \text{Upsample}(P_{i+1}) \tag{2}$$

By combining deep and shallow information, the decoder can benefit from both their semantic richness and spatial precision.

F. Attention Gate

The attention gate on each skip connection generates a soft spatial attention map:

$$\alpha = \sigma(\mathbf{W}_\psi \cdot \text{ReLU}(\mathbf{W}_g\mathbf{g} + \mathbf{W}_x\mathbf{x} + \mathbf{b}_g)) \tag{3}$$

where $\mathbf{g}$ is the gating signal which comes from the decoder, $\mathbf{x}$ is the encoder's skip feature, while σ represents the sigmoid activation. Multiplying the Skip feature by α suppresses unimportant background activations.

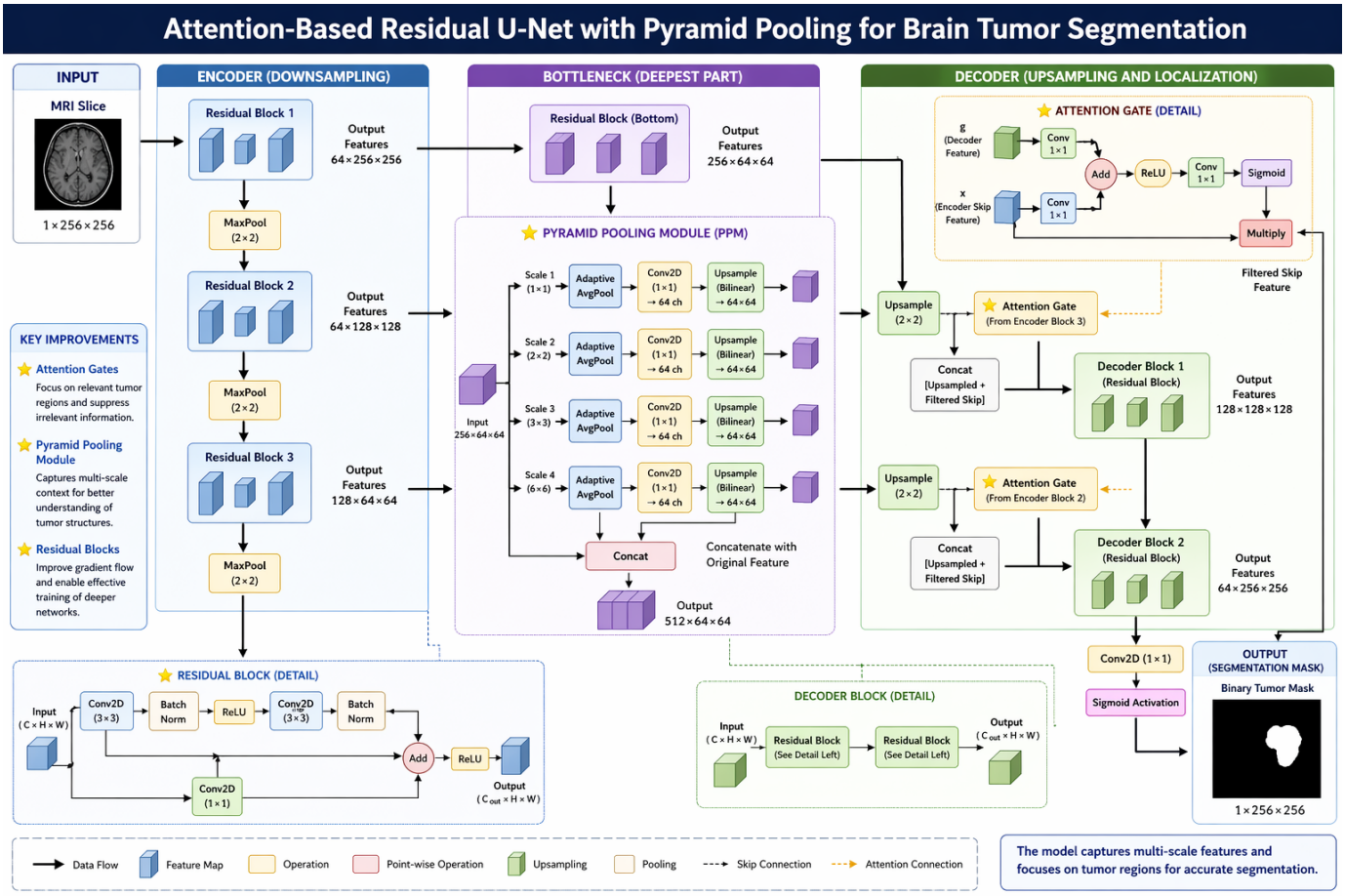

Fig. 1. Proposed system architecture

G. Pyramid Pooling Module

The bottleneck feature map $\mathbf{F}$ is pooled at scales $\{1, 2, 3, 6\}$, upsampled back to the original size, and concatenated with $\mathbf{F}$:

$$\mathbf{F}_{\text{ppm}} = \text{Conv}([\mathbf{F};\ \text{Up}(P_1);\ \text{Up}(P_2);\ \text{Up}(P_3);\ \text{Up}(P_6)]) \quad (4)$$

This improves the bottleneck representation by adding local texture and global contextual information.

H. Hybrid Loss Function

A hybrid loss was used to handle both region overlap and hard-pixel supervision:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{Dice}} + \lambda_2 \mathcal{L}_{\text{BCE}} + \lambda_3 \mathcal{L}_{\text{Focal}} \quad (5)$$

- **Dice Loss** ($\mathcal{L}_{\text{Dice}}$): Optimizes volumetric overlap between expected and ground-truth masks.
- **Binary Cross-Entropy** ($\mathcal{L}_{\text{BCE}}$): offers pixel-level supervision for all spatial locations.
- **Focal Loss** ($\mathcal{L}_{\text{Focal}}$): The algorithm prioritizes learning from tough and minority tumor pixels over easy negative pixels.

I. Training Configuration

Table II sums up the experimental training setup.

TABLE II
TRAINING CONFIGURATION

Parameter	Value
Framework	PyTorch
Platform	Google Colab (GPU)
Optimizer	Adam
Batch Size	8
Epochs	30
Input Resolution	256×256
Task Type	Binary Segmentation

V. EVALUATION METRICS

Performance was evaluated using standard confusion matrix statistics based on the number of true positives (TP), true negatives (TN), false positives (FP), and false negatives:

$$\text{Dice} = \frac{2\,\text{TP}}{2\,\text{TP} + \text{FP} + \text{FN}} \tag{6}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{7}$$

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{8}$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{9}$$

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{10}$$

The Dice coefficient (6) is primary metric for segmentation evaluates the overlap between the predicted mask and the ground truth. Sensitivity identifies tumor pixels correctly, but specificity protects healthy tissue from incorrect classification.

VI. Results and Discussion

A. Final Model Performance

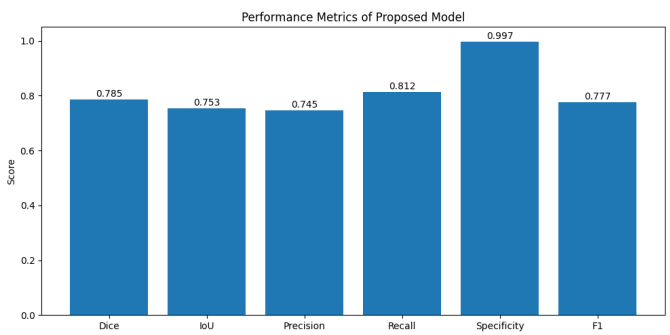

Fig. 2. Segmentation Performance Metrics of Proposed Model

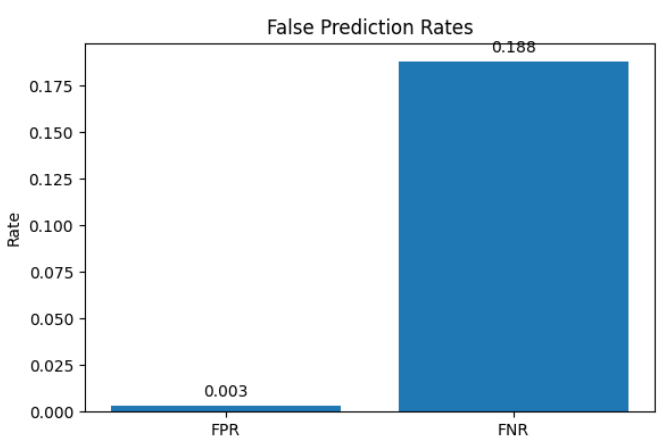

Fig. 3. False Positive Rate and False Negative Rate

Table III evaluates the proposed model's performance on the LGG MRI validation set.

TABLE III
Final Model Performance on the LGG MRI Validation Set.

Metric	Value
Accuracy	0.995
Dice Score	0.839
Sensitivity	0.781
Specificity	0.997
False Positives	Very Low
Boundary Quality	Good

B. Discussion

The Dice score of 0.839 suggests significant volumetric overlap between the predicted masks and ground truth annotations. The sensitivity of 0.781 indicates that most tumor pixels were effectively recognized, while the specificity of 0.997 shows that healthy tissue was rarely misclassified as tumor. The overall accuracy of 0.995 indicates strong pixel-wise categorization performance over the entire image.

Compared to simpler baseline structures, the predicted masks were cleaner, less noisy, and more constant around tumor boundaries. Residual blocks sustained training convergence, FPN increased tiny tumor detection, and attention gates reduced spurious activations in background tissue. The pyramid pooling module enhanced context awareness, allowing the model to handle tumors with changing spatial extent.

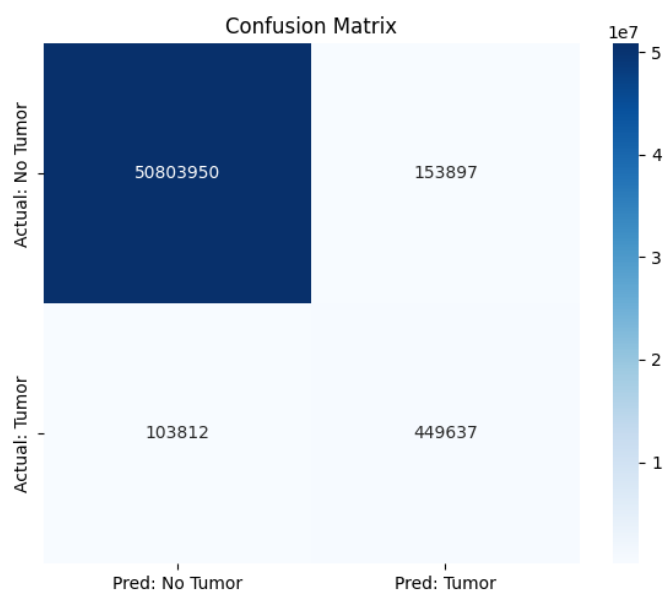

Fig. 4. Pixel-Level Confusion Matrix of Proposed Model

VII. Comparative Analysis

A. Qualitative Comparison

Table IV outlines the qualitative aspects of each examined architecture.

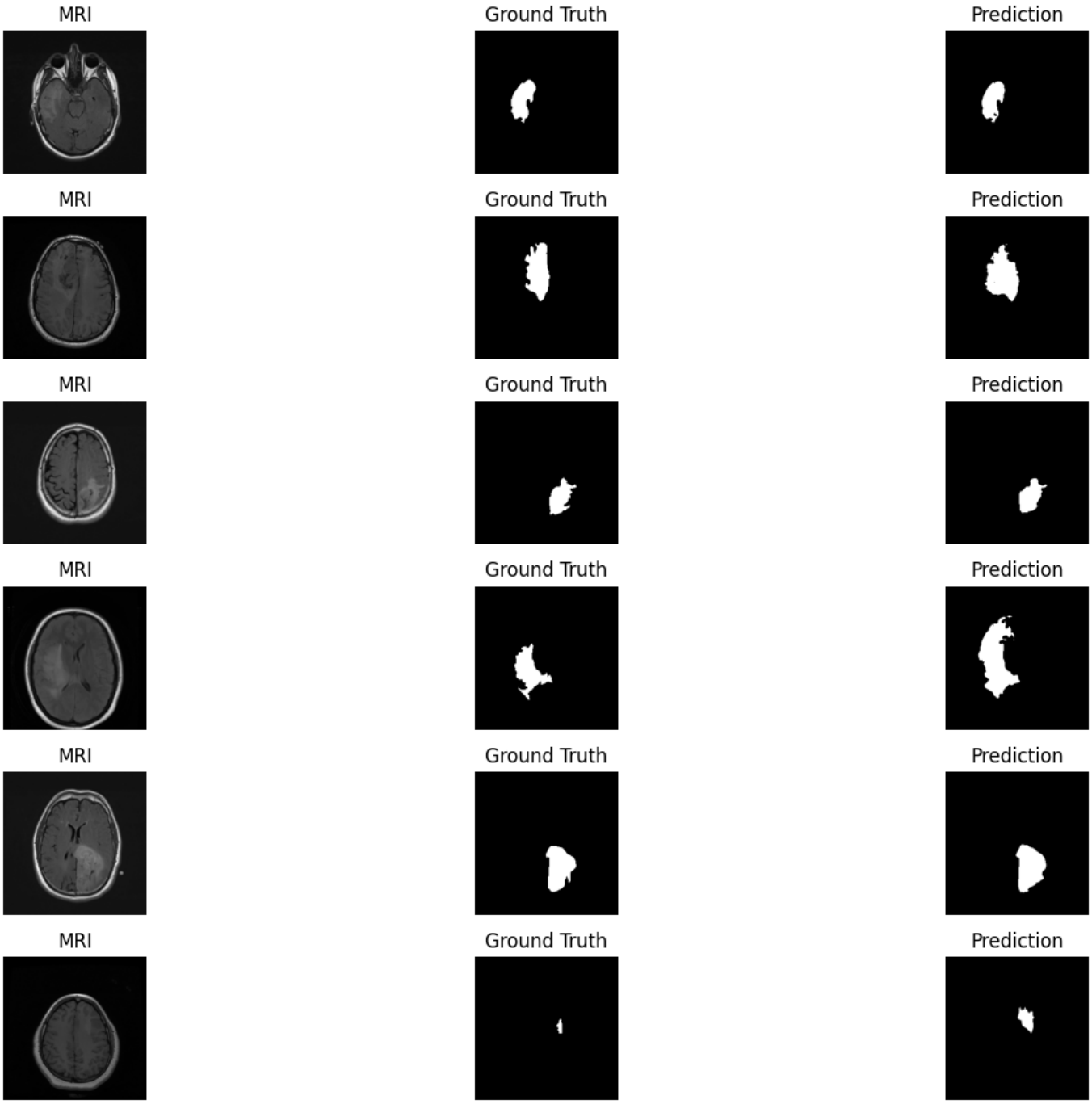

Fig. 5. Qualitative Results: MRI Input, Ground Truth Mask, and Predicted Mask

TABLE IV
QUALITATIVE ARCHITECTURAL COMPARISON

Model	Main Idea	Limitation	Performance
Standard U-Net	Encoder-decoder segmentation	Misses small tumors	Moderate
Attention U-Net	Focus on relevant regions	Limited scale awareness	Better
U-Net + Pyramid Pooling	Multi-scale context	Some false positives remain	Better
Proposed Model	Residual + FPN + Attention + Hybrid	Higher training cost	Best

B. Quantitative Comparison

Table V enables a direct numerical comparison of all examined models.

TABLE V
NUMERICAL PERFORMANCE COMPARISON

Metric	Standard U-Net	Attention U-Net	Proposed
Accuracy	0.991	0.993	**0.995**
Dice Score	0.740	0.790	**0.839**
Sensitivity	0.710	0.760	**0.781**
Specificity	0.980	0.990	**0.997**
False Positives	Medium	Low	Very Low
Boundary Quality	Average	Good	Best

The suggested architecture outperformed all baseline models, showing significant increases in Dice score (+0.099 over conventional U-Net), specificity, and boundary reconstruction quality.

VIII. APPLICATIONS AND LIMITATIONS

A. Clinical Applications

The created system could be used in various clinical contexts:

- Computer-aided diagnosis assistance for radiologists
- Pre-surgical tumor localization and margin delineation
- Radiotherapy target volume planning
- Longitudinal tumor growth monitoring
- Clinical training and education systems
- Remote healthcare help in resource-constrained environments

B. Limitations

Despite solid quantitative results, the existing method has many limitations:

- The model was trained using a modest single-domain dataset.
- Processing works with 2D slices rather than whole volumetric 3D data.
- Performance may suffer across different MRI scanner protocols and field strengths.
- GPU resources are necessary for both practical training and inference throughput.

IX. FUTURE SCOPE

Several approaches are intended to strengthen the segmentation system:

- **Larger Datasets:** Training on the BraTS multi-institutional benchmark can help enhance generalisation across scannner types.
- **3D Volumetric Models:** Extending the architecture to full 3D U-Net or other volumetric approaches.
- **Transformer Integration:** Incorporating vision transformer encoders and improving capturing global context.
- **Semi-Supervised Learning:** Leveraging unlabeled data to reduce dependence on expensive pixel-level matching.
- **Explainability:** To improve clinical interpretability.
- **Multi-Class Segmentation:** Extending to segmentation of multiple tumor subregions.
- **Clinical Deployment:** Deploying the system as a web service for hospitals.

X. CONCLUSION

This research introduces a strong brain tumor segmentation system based on U-Net and augmented with residual learning, Feature Pyramid Network, attention mechanisms, pyramid pooling, and hybrid loss optimization. The suggested approach tackles frequent failure modes in current segmentation designs, including missed tiny tumors, noisy predictions, and inadequate border reconstruction. The LGG MRI Segmentation dataset showed high performance (Accuracy = 0.995, Dice Score = 0.839, Specificity = 0.997), indicating the efficiency of the overall architectural upgrades.

This project taught real skills in deep learning pipeline design, medical picture preprocessing, metric-driven model evaluation, and iterative architecture improvement. Combining complementing architectural approaches in a single framework leads to accurate and clinically useful segmentation systems, which can serve as the foundation for computer-aided diagnosis tools.

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